

Roadblock cluster triage — raw clusters under final mechanism umbrellas (detailed)

Source: `bench/dataset.jsonl` (627 rows over 558 distinct branches; 398 validated / 229 inconclusive). 57 raw clusters merged into 19 final features. Each shaded block is a final feature (umbrella); rows beneath are the raw clusters (per decisive shape) merged into it.

Raw cluster	Shape (res/blk)	#br	Cluster description	Tool	Metric(s)	Metric meaning / prediction	Verify rule
[I2S-pro] i2s_string_literal_substitution (final feature, n=185)							
i2s_exact_literal_gate	LWL (vpc/vp+naive)	67	vpc resolves an exact-literal equality gate because its I2SRandReplace splices the logged comparand literal into a fixed input offset, which value_profile's Hamming/CMP_MAP gradient cannot land	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich > 0 (strictly): the exact target literal at the gate offset is MORE frequent in the vpc-resolving corpus than in the vp/naive-blocking corpora, because only the I2S-bearing winner can place the all-or-nothing constant. An ALTERNATIVE mechanism predicts differently: if vpc resol...	signed_target_enrich >= 1.0
i2s_fixed_offset_literal_value_gate	WLWL (cmp+vpc/vp+naive)	67	cmplog (I2S) substitutes a logged CMP literal at a fixed gate offset, so its corpus is enriched in the target value at that offset relative to naive; value_profile/naive cannot synthesize the exact literal	operand_enrichment	signed_target_enrich, decoy_enrich	For a fixed-offset literal value gate, the cmplog corpus is ENRICHED in the target literal at the gate offset relative to naive: signed_target_enrich > 0 (i2s-PRO direction). A pure-coverage / value-profile-only mechanism predicts NO cmplog-specific enrichment at the operand offset (signed_target.e...	signed_target_enrich >= 0.8
i2s_fixed_offset_literal_substitution	W_WL (cmp+vpc/naive)	20	I2S enriches the cmplog corpus with the exact CMP-target byte(s) at the gate offset (literal substitution), which naive's blind havoc cannot place	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich > 0: the cmplog corpus is log2-enriched (vs naive) for the exact target byte value at the fixed gate offset. DISCRIMINATING because an i2s-pro literal-substitution mechanism deterministically plants the CMP operand at that one offset, so its corpus concentrates on the target va...	signed_target_enrich >= 1.0
i2s_planted_anchor_literal_presence_scaffold_gate	WLWL (cmp+vpc/vp+naive)	10	I2S splices one or more logged CMP anchor literals (URL scheme '://', 'HTTP'/'HTTP/2', header names, protocol tokens 'pop3', chunk-type FOURCCs 'IDAT'/'sRGB'/'eXif'/'sCAL'/'tIME', OpenType table tags 'GPOS'/'GSUB', the Mach-O ma...	token_count	winner_literal_count, loser_literal_count, literal_pre...	If the branch is reached via an I2S-planted anchor literal, that literal is PRESENT in the cmplog-resolving seeds and DEPLETED in the naive/value_profile-blocking seeds: winner_literal_count >= 1 AND literal_presence_ratio << 1. The alternative this rules out is that value_profile's Hamming gradien...	winner_literal_count >= 1.0 AND literal_presence_ratio <= 0.34
literal_value_gate_cmp_solvable	WWWL (cmp+vp+vpc/naive)	7	A fixed-offset literal value gate (FOURCC/tag/enum/single byte) is solved by any CMP-aware fuzzer (I2S substitution or VP Hamming gradient) while naive's blind havoc cannot place the literal	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich >= +1.0 (CMP-aware corpora over-represent the target literal bytes at the gate offset relative to naive) AND decoy_enrich <= 0 (CMP-aware does not over-represent the off-gate decoy bytes). DISCRIMINATING: a pure structural/length mechanism (not a literal-value gate) would predi...	signed_target_enrich >= 1.0 AND decoy_enrich <= 0.0
parallel_fixed_offset_literal_gate	WW_L (cmp+vp/naive)	6	A fixed-offset 1-4 byte VALUE literal/enum gates the branch; both I2S (direct substitution) and VP (Hamming/prefix gradient) drive the gate window toward the target value AND away from the decoy, so BOTH resolving corpora are enr...	operand_enrichment	signed_target_enrich, decoy_enrich, vp_signed_target_e...	Each resolving fuzzer's corpus drives the gate window toward the target literal: EITHER it over-represents the target byte vs naive (signed_target_enrich > 0.5) OR it depletes the decoy byte vs naive (decoy_enrich < -1.0) — the two are complementary corpus-content footprints of the same drive (targ...	((cmplog.signed_target_enrich > 0.5) or (decoy_enrich < -1.0)) and ((value_profile.signed_target_enrich > 0.5) or (vp.d...))

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i2s_literal_operand_substitution	WLW_ (cmp+vpc/vp)	3	I2SRandReplace splices the exact logged comparand literal into a fixed head gate offset; VP's CMP_MAP Hamming gradient cannot reach the exact value within budget => cmplog corpus enriched in the target byte vs naive	operand_enrichment	signed_target_enrich, decoy_enrich	i2s-pro => signed_target_enrich >= 1.0 : the cmplog corpus carries the exact target literal byte at the fixed gate offset at a higher rate than naive, because I2SRandReplace plants the logged comparand there and VP/havoc cannot. A 'VP-gradient-was-sufficient' or 'literal-is-corpus-ambient' alternat...	signed_target_enrich >= 1.0
i2s_relational_collision_gate	W_L (cmp/naive)	3	I2S pastes an exact short literal (an XML entity name like amp/apos, or the comment marker <!--) at a content-dependent position rather than a fixed file offset — mechanistically a string-literal substitution, FOLDED into i2s_string_literal_substitution; confirmed by Layer-1 cmplog>naive + absence of a byte-signature.	operand_enrichment	gate_signature	operand.enrichment finds NO fixed input offset that gates the branch (gate_signature == false): the literal sits at a grammar-determined position, so fixed-offset enrichment is not measurable; the cmplog>naive Layer-1 advantage places it as the floating-offset variant of literal substitution.	gate_signature == false
i2s_exact_literal_constant_gate	WL_L (cmp/vp+naive)	1	cmplog plants an exact fixed-offset literal/magic constant the VP gradient cannot climb to	operand_enrichment	signed_target_enrich, decoy_enrich	cmplog's saved corpus is enriched in the TARGET literal byte(s) at the gate offset relative to naive (signed_target_enrich > 0), because I2SRandReplace plants the intercepted constant. A pure value-profile gradient mechanism would instead drive the byte toward the target by Hamming distance and wou...	signed_target_enrich >= 1.0 AND decoy_enrich <= 0.0
i2s_literal_substitution_gate	W_L (cmp/naive)	1	A fixed-offset literal CMP gate that I2S satisfies by splicing the logged operand byte(s) into the input; cmplog's corpus is enriched for the target literal at the gate offset where naive's is not	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich >= 1.0 (cmplog's saved corpus carries the target literal at the gate offset >=2x more than naive's, i.e. log2 enrichment positive and substantial) AND the gate has a byte-traceable fixed-offset operand (operand.enrichment returns a gate signature rather than no_gate_signature)....	gate_signature == true AND signed_target_enrich >= 1.0
[I2S-pro] i2s_numeric_tag_substitution (final feature, n=39)							
i2s_fixed_offset_literal_substitution	W_WL (cmp+vpc/naive)	17	I2S enriches the cmplog corpus with the exact CMP-target byte(s) at the gate offset (literal substitution), which naive's blind havoc cannot place	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich > 0: the cmplog corpus is log2-enriched (vs naive) for the exact target byte value at the fixed gate offset. DISCRIMINATING because an i2s-pro literal-substitution mechanism deterministically plants the CMP operand at that one offset, so its corpus concentrates on the target va...	signed_target_enrich >= 1.0
i2s_exact_literal_gate	_LWL (vpc/vp+naive)	11	vpc resolves an exact-literal equality gate because its I2SRandReplace splices the logged comparand literal into a fixed input offset, which value_profile's Hamming/CMP_MAP gradient cannot land	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich > 0 (strictly): the exact target literal at the gate offset is MORE frequent in the vpc-resolving corpus than in the vp/naive-blocking corpora, because only the I2S-bearing winner can place the all-or-nothing constant. An ALTERNATIVE mechanism predicts differently: if vpc resol...	signed_target_enrich >= 1.0
i2s_fixed_offset_literal_value_gate	WLWL (cmp+vpc/vp+naive)	6	cmplog (I2S) substitutes a logged CMP literal at a fixed gate offset, so its corpus is enriched in the target value at that offset relative to naive; value_profile/naive cannot synthesize the exact literal	operand_enrichment	signed_target_enrich, decoy_enrich	For a fixed-offset literal value gate, the cmplog corpus is ENRICHED in the target literal at the gate offset relative to naive: signed_target_enrich > 0 (i2s-PRO direction). A pure-coverage / value-profile-only mechanism predicts NO cmplog-specific enrichment at the operand offset (signed_target_e...	signed_target_enrich >= 0.8

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i2s_exact_literal_constant_gate	WL_L (cmp/vp+naive)	5	cmplog plants an exact fixed-offset literal/magic constant the VP gradient cannot climb to	operand_enrichment	signed_target_enrich, decoy_enrich	cmplog's saved corpus is enriched in the TARGET literal byte(s) at the gate offset relative to naive (signed_target_enrich > 0), because I2SRandReplace plants the intercepted constant. A pure value_profile gradient mechanism would instead drive the byte toward the target by Hamming distance and wou...	signed_target_enrich >= 1.0 AND decoy_enrich <= 0.0
[I2S-pro] i2s_structural_assembly_reach_depth (final feature, n=8)							
i2s_structural_assembly_reach_depth	W_WL (cmp+vpc/naive)	6	I2S repairs interdependent multi-field structure (ELF section table / PNG iTXt chunk grammar / XML scheme+attribute parse) so the winner corpus drives the gated loop/branch line DEEPER per seed than the loser, rather than flippin...	depth_reach	depth_lift, winner_depth, loser_depth, winner_deeper	depth_lift > 1 AND winner_deeper=true: the winner-resolving seeds execute the gated loop/branch line MORE per seed than the loser-blocking seeds, because I2S repaired enough of the surrounding multi-field structure for the parser to advance through it. DISCRIMINATING: a pure single-literal-substitu...	depth_lift >= 1.5 AND winner_deeper == true
i2s_structural_depth_gate	WLWL (cmp+vpc/vp+naive)	2	I2S assembles a MULTI-FIELD structure whose downstream effect is that the cmplog-RESOLVING corpus, run dynamically, DRIVES execution deeper to the blocker line AND TAKES the decisive side, while the naive/value_profile-BLOCKING c...	depth_reach	depth_lift, side_lift, winner_depth, loser_depth, winn...	If I2S unlocks the branch by assembling a multi-field structure, the cmplog-resolving corpus run dynamically REACHES the blocker line far more per seed AND TAKES the decisive side, while the naive-blocking corpus stalls: depth_lift >= 2 AND side_lift >= 3. The ALTERNATIVE ruled out: 'the loser corp...	depth_lift >= 2.0 AND side_lift >= 3.0 AND winner_deeper == true
[I2S-pro] i2s_operand_value_precision (final feature, n=6)							
i2s_operand_value_precision_at_equal_reach_gate	WLWL (cmp+vpc/vp+naive)	6	NEW (round 3). The blocking arm (naive AND, critically, value_profile — which has the CMP_MAP feedback but NO operand-substitution mutator) REACHES the gate line just as deeply as the winner (often deeper: the cards show losers h...	value_distance_reached	cmp_min_distance, vpc_min_distance, vp_min_distance, n...	If I2S unlocks the branch by substituting the exact operand at a line both arms already reach, the cmplog/vpc winner lands residual ~0 (operand present) while the closest value_profile/naive loser stalls at a positive residual: winner_closer == true AND distance_gap > 0 AND cmp_min_distance is at/n...	winner_closer == true AND distance_gap > 0 AND distance_closure_ratio >= 0.5
[I2S-anti] i2s_anti_target_depletion (final feature, n=25)							
i2s_decoy_substitution_target_depletion	LW_W (vp+naive/cmp)	23	I2S splices a WRONG/competing but abundant CMP operand into the input (a well-known font-table tag, a script-tag/codepoint constant, an in-range fixed-point magnitude, or a competing protocol value), pinning the cmplog corpus ont...	operand_enrichment	signed_target_enrich, decoy_enrich	On the gate offset the cmplog corpus is DEPLETED of the target value relative to naive (signed_target_enrich < -0.3) AND/OR ENRICHED in the decoy operand (decoy_enrich > 0): I2S is actively steering bytes AWAY from the winner value. A pure-overhead mechanism (I2S has no operand to substitute) would...	skip != 'no_gate_signature' AND signed_target_enrich < -0.3
i2s_energy_diversion_decoy_pin	L_W (naive/cmp)	2	the gate is an ungradable fixed-offset equality; plain havoc places the literal by random mutation while cmplog diverts energy and leaves its corpus depleted of (pinned away from) the target value at the gate offset	operand_enrichment	signed_target_enrich, decoy_enrich	cmplog's corpus is DEPLETED of the target value 0x05E9 at the gate offset relative to naive: signed_target_enrich < -0.3 (anti). I2S introspection diverts energy without learning the equality, so cmplog under-represents the literal that havoc reaches incidentally. The PRO alternative (I2S substitut...	signed_target_enrich < -0.3
[I2S-anti] i2s_anti_decoy_overfit (final feature, n=14)							

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decoy_overcommit_wrong_literal	_WLW (vp+naive/vpc)	6	vpc's I2S substitution pins the corpus to a WRONG decoy literal (sibling case / enumerated case / valid-range boundary), flooding the loser corpus with a value that blocks the target arm	operand_enrichment	decoy_enrich, signed_target_enrich	vpc's corpus is ENRICHED for the decoy literal relative to vp (decoy_enrich > 0) while NOT enriched for the target value (signed_target_enrich <= 0). G1-discriminating: pure overhead/throughput drain (the rival hypothesis) predicts NO directional byte-frequency difference between vpc and vp corpora...	decoy_enrich > 0.8 AND signed_target_enrich < 0.3
i2s_decoy_substitution_overfit	LWLW (vp+naive/cmp+vpc)	6	I2S substitutes a fixed-offset structural literal (decoy) that pins the loser corpus to the valid-structure path and away from the winner's free-byte target	operand_enrichment	signed_target_enrich, decoy_enrich	On the I2S-bearing (cmplog/vpc) corpus vs naive corpus, the decoy structural literal is ENRICHED (decoy_enrich > 0) while the winner-target byte is DEPLETED (signed_target_enrich < 0). A pro-I2S 'magic-gate' alternative would predict the opposite (signed_target_enrich > 0, the I2S arm enriches the ...	signed_target_enrich < -0.3
i2s_case_constant_decoy_substitution	L_LW (naive/cmp+vpc)	1	I2S keeps copying enumerated switch case-constants (decoys) into the option-selector byte, pinning control onto a matching case and depleting cmplog/vpc corpora of the non-enumerated values that fall through to default; plain hav...	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich < 0 (cmplog/vpc corpus is DEPLETED of the default-reaching non-enumerated value relative to naive) AND decoy_enrich > 0 (cmplog/vpc corpus is ENRICHED for the enumerated case-constant). An ALTERNATIVE in which I2S played no role (e.g. a pure corpus-scale or structural-assembly ...	signed_target_enrich < -0.3
i2s_diverts_from_exact_codepoint_literal	LWLW (vp+naive/cmp+vpc)	1	Gate is an EXACT-equality test against a short codepoint literal that the WINNER (non-I2S) corpus lands by free havoc byte mutation, while the I2S stage diverts mutation energy away from the literal so the I2S-arm corpus stalls F...	value_distance_reached	vp_min_distance, vpc_min_distance, cmp_min_distance, n...	The WINNER arm (naive/value_profile) corpus contains a 2-byte window at Hamming distance 0 to 0x0E0E/0x0E0F (it actually lands the literal), while the LOSER arm (cmplog/vpc) stalls at a strictly larger min distance (distance_gap > 0 with winner_closer=true). This rules out the NULL alternative 'I2...	winner_closer == true AND distance_gap >= 0.25 AND naive_min_distance <= 0.0
[I2S-anti] i2s_anti_structural_byte_corruption (final feature, n=10)							
i2s_anti_structural_byte_corruption	LWWW (vp+vpc+naive/cmp)	7	I2S RandReplace overwrites the exact structural/category/literal bytes naive havoc assembles, depleting (not enriching) the target byte in cmplog's corpus relative to naive	operand_enrichment	signed_target_enrich	If I2S is ACTIVELY CORRUPTING the structural byte, cmplog's corpus is byte-DEPLETED at the target offset vs naive => signed_target_enrich < 0 (anti). An I2S-IRRELEVANT gate (the no-preimage class) returns no_gate_signature; a pro-I2S gate returns > 0. A clearly NEGATIVE enrichment discriminates act...	signed_target_enrich < -0.3
vpc_corpus_shallower_context_reach	_WLW (vp+naive/vpc)	2	vpc's I2S substitution pulls the loser corpus toward shallow / degenerate inputs that never DRIVE THE CALL CHAIN to the deep shaping context where the gate is dispatched; vp's gradient-grown corpus reaches that context far more p...	depth_reach	depth_lift, winner_depth, loser_depth, winner_deeper	vp's resolving seeds execute the gate-line context substantially more per seed than vpc's blocking seeds (winner_deeper == true AND depth_lift >= 1.5), i.e. vpc-blocking seeds rarely even reach the switch dispatch context. G1-discriminating: the two PRIOR mechanisms already tried-and-failed on 5623...	winner_deeper == true AND depth_lift >= 1.5

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vpc_anti_sibling_subtree_depth_diversion	LWL_ (vp/cmp+vpc)	1	The I2S liability in vpc manifests as EXECUTION-DEPTH diversion at the target branch line: the vpc-BLOCKING corpus, having spliced the gate's logged comparands, gets funneled into the SIBLING / true-side state-machine sub-tree (6...	depth_reach	winner_depth, loser_depth, depth_lift, side_lift, winn...	If the vpc regression is an EXECUTION-DEPTH diversion (I2S funnels the vpc corpus into the sibling/true-side sub-tree, away from the target branch line), the vp-RESOLVING seeds execute the target branch line markedly MORE per seed than the vpc-BLOCKING seeds: winner_depth > loser_depth, depth_lift ...	winner_deeper == true AND depth_lift >= 1.5 AND winner_depth > loser_depth
[VP-pro] vp_gradient_value_distance_closure (final feature, n=28)							
cmp_gradient_closes_to_operand	WWWL (cmp+vp+vpc/naive)	8	The CMP-aware corpus (value_profile Hamming gradient / I2S substitution) lands at or near the gate operand literal in a sliding-window distance sense, while naive's corpus stays far from it – a CLOSENESS signal that survives dis...	value_distance_reached	vp_min_distance, cmp_min_distance, naive_min_distance,...	winner_closer == true AND distance_gap >= 0.25 – the CMP-aware arm's corpus reaches a sliding window whose Hamming distance to the operand literal is materially smaller than naive's nearest window. DISCRIMINATING: a NON-CMP mechanism (scheduling / energy / pure structural-length) predicts distance...	winner_closer == true AND distance_gap >= 0.25
vp_cmpmap_distance_closure_fixed_operand	_WWWL (vp+vpc/naive)	5	For gates with a SINGLE fixed comparison operand, value_profile's CMP_MAP gradient admits a corpus that lands at a strictly smaller residual Hamming/prefix distance to that operand than the naive-blocking corpus ever reaches. (CA...	value_distance_reached	vp_min_distance, naive_min_distance, distance_closure....	naive_min_distance > 0 (the naive-blocking corpus systematically STALLS at a positive residual to the fixed operand – carries 'RGB ' not the FOURCC, no '—') while value_profile closes it, so distance_closure_ratio is large. DISCRIMINATING vs a no-gradient null (VP got the operand by the same blin...	naive_min_distance > 0 AND distance_closure_ratio >= 0.5
vp_gradient_value_distance_climb	L_WL (vpc/cmp+naive)	5	vpc's CMP_MAP gradient ratchets the corpus byte-by-byte toward a CONCRETE literal token that cmplog's edge-only signal cannot bank near-miss progress toward	value_distance_reached	w_min_hamming_to_literal, l_min_hamming_to_literal, di...	W seeds sit measurably CLOSER to the gate literal than L seeds (distance_gap > 0 AND w_min_hamming_to_literal low), because the CMP_MAP gradient preserved near-miss seeds inching toward the literal. This is DISCRIMINATING: the competing 'I2S substitution alone suffices' alternative predicts cmplog...	distance_gap >= 0.25 AND w_min_hamming_to_literal <= 0.30
vp_basin_escape	_W_L (vp/naive)	3	VP CMP_MAP steers AWAY from a dominant magic-value basin: there is no single target literal – the true side wants ANY value but the corpus-dominant magic. VP-resolving seeds DEPART the basin (multi-modal: spread over several non-...	value_distance_reached	naive_basin_lock_fraction, vp_basin_lock_fraction, vp_...	G1: naive-blocking seeds are basin-locked (high naive_basin_lock_fraction) while VP-resolving seeds have left the basin (low vp_basin_lock_fraction) AND spread over MULTIPLE distinct non-magic prefixes (vp_value_modality >= 3). The direct-literal alternative predicts the opposite signature: VP seed...	basin_departure_lift > 0 AND naive_basin_lock_fraction >= 0.7 AND vp_value_modality >= 3
parallel_hamming_gradient_value_closure	WW_L (cmp+vp/naive)	2	NEW (confirmed round 2). A fixed-width VALUE operand (a 2-byte enum constant) gates the branch, but the discriminating corpus-content signal is not byte-frequency at one offset — it is HAMMING-DISTANCE CLOSURE: both resolving arm...	value_distance_reached	vp_min_distance, cmp_min_distance, naive_min_distance,...	Both resolving arms land Hamming-close to 0x003e and materially closer than naive: vp_min_distance and cmp_min_distance are near 0 while naive_min_distance is larger, giving distance_gap > 0 AND distance_closure_ratio high. DISCRIMINATING: (a) if ONLY cmplog closes and VP stays at naive's distance,...	winner_closer == true and distance_gap >= 0.25 and vp_min_distance <= 0.25 and cmp_min_distance <= 0.25
vp_climbs_numeric_magnitude_to_threshold	LW_L (vp/cmp+naive)	2	value_profile's gradient drives a parsed numeric field MONOTONICALLY toward / across an inequality threshold via CMP_MAP operand-distance buckets on the integer comparison, while naive (edge-only) and cmplog (no single constant t...	value_distance_reached	vp_min_distance, naive_min_distance, distance_gap, dis...	winner_closer == true AND distance_gap > 0. DISCRIMINATING vs literal substitution: an inequality gate has NO single target constant for I2S to splice. DISCRIMINATING vs naive luck: card reports naive T=0 F=10 (never crossed).	winner_closer == true AND distance_gap > 0

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vp_gradient_distance_climb	LWLL (vp/cmp+vpc+naive)	1	value_profile's CMP_MAP gradient climbs a comparison-operand distance ramp toward a structural/syntactic gate state that cmplog- and naive-blocking seeds never approach	value_distance_reached	vp_min_distance, naive_min_distance, cmp_min_distance,...	G1: value_profile's resolving seed set reaches a STRICTLY CLOSER minimum CMP-operand distance to the gate than BOTH cmplog's and naive's blocking seed sets (vp_min_distance < cmplog_min_distance AND vp_min_distance < naive_min_distance), and satisfies a larger fraction of gate CMPs (vp_gate_cmp_sat...	vp_min_distance < cmplog_min_distance AND vp_min_distance < naive_min_distance AND vp_gate_cmp_satisfied_frac >= naive...
vp_gradient_scalar_operand_distance	LWWL (vp+vpc/cmp+naive)	1	value_profile's CMP_MAP gradient lands its corpus closer to a FIXED scalar gate operand than cmplog/naive get – the genuine value-distance subtype, restricted to the one gate that has an extractable contiguous literal AND on-dis...	value_distance_reached	vp_min_distance, vpc_min_distance, cmp_min_distance, n...	G1: if VP's gradient drives the corpus onto the '<<' shift operator that gates OP_ShiftLeft, the vp/vpc resolving seeds land ON or NEAR '<<' (winner_min ~ 0) while the cmplog/naive blocking seeds stall away from it (distance_gap > 0, winner_closer true). ALTERNATIVES ruled out: (a) plain coverage l...	winner_closer == true AND distance_gap >= 0.25 AND cmp_min_distance >= 0.40
vp_pro_cmpmap_gradient_rescue	LWWW (vp+vpc+naive/cmp)	1	VP CMP_MAP Hamming/prefix-distance gradient gives graded partial-progress feedback toward the gate comparand, letting value_profile/vpc retain near-miss inputs that approach the target while cmplog's edge-only binary feedback sta...	value_distance_reached	vp_min_distance, cmp_min_distance, distance_gap, winne...	If VP's gradient is the rescue mechanism, value_profile's corpus REACHES strictly closer to the gate comparand than cmplog's (vp_min_distance < cmp_min_distance, winner_closer=true) while cmplog plateaus far from the target. If both reached equal distance, the rescue is NOT gradient-driven (refuted...	winner_closer == true AND distance_gap >= 0.15
[VP-pro] vp_gradient_drives_assembly_depth (final feature, n=15)							
vp_gradient_structural_depth	LWWL (vp+vpc/cmp+naive)	5	value_profile's gradient-built corpus reaches a DEEPER parse/codegen state along the gating call-chain than cmplog/naive – the structural/derived/grammar subtype where the gate is a parse DEPTH, not a scalar operand	depth_reach	depth_lift, side_lift, winner_depth, loser_depth, winn...	G1: if VP's gradient climbs the parse chain, value_profile's resolving corpus drives the blocker line far deeper / takes the blocked side, while naive's blocking corpus stalls shallow (depth_lift >> 1, winner_deeper true, side_lift > 1). ALTERNATIVE ruled out: 'no real gradient – VP resolved by a ...	winner_deeper == true AND depth_lift >= 1.5 AND side_lift >= 1.5
vp_cmpmap_iteration_depth_reach	_WWL (vp+vpc/naive)	3	NEW (round-3 pivot). For RANGE / LOOP / DEEP-DISPATCH gates with NO harvestable single fixed-byte operand (digit-range loops past a counter threshold, a hex-charref loop reaching its non-hex terminator, a deep codepoint switch ar...	depth_reach	depth_lift, winner_depth, loser_depth, winner_deeper	winner_depth > loser_depth with depth_lift >= 1.5 : the value_profile-resolving corpus drives the gate's branch line to a materially greater per-seed execution depth than the naive-blocking corpus, which stalls shallow (naive's digit run never reaches the loop-counter threshold; naive's input never...	depth_lift >= 1.5 AND winner_deeper
vp_drives_corpus_deeper_into_assembly	LWW_ (vp+vpc/cmp)	2	for the STATE-MACHINE / RECURSIVE-ASSEMBLY gates, value_profile's CMP_MAP gradient admits corpus members that DYNAMICALLY REACH a deeper point in the parse/recursion than the cmplog-blocking corpus – the winner seeds drive the g...	depth_reach	depth_lift, winner_depth, loser_depth, winner_deeper	depth_lift = winner_line_count / loser_line_count >= 2.0 AND winner_deeper : the VPC-resolving corpus reaches the gate line at least 2x more per seed than the cmplog-blocking corpus. DISCRIMINATING vs round-0 (depth is a runtime reach count at the gate line, not a Hamming residual) and vs a pure-I2...	depth_lift >= 2.0 AND winner_deeper
vp_drives_gate_deeper_per_seed	LWLL (vp/cmp+vpc+naive)	2	value_profile's CMP_MAP gradient admits corpus members that REACH the gate region deeper / more times PER SEED than naive's blocking seeds - the gate sits behind a parse/VDBE/shaping loop and VP's structured seeds re-enter and ex...	depth_reach	depth_lift, side_lift, winner_depth, loser_depth, winn...	G1: value_profile's RESOLVING seed set drives the gate line deeper PER SEED than naive's BLOCKING seed set (depth_lift >= 1.5 AND winner_deeper == true), i.e. VP's seeds reach and re-exercise the gate region while naive's short fragments die before / at the gate's containing loop. DISCRIMINATING: t...	depth_lift >= 1.5 AND winner_deeper == true

Raw cluster	Shape (res/blk)	#br	Cluster description	Tool	Metric(s)	Metric meaning / prediction	Verify rule
vp_corpus_drives_sanitizes_loop_deeper	L_W_ (vpc/cmp)	1	value_profile's CMP_MAP gradient retains a corpus that drives the GATE LINE through MORE iterations / deeper into the loop per seed – the VPC-resolving seeds re-enter the per-record sanitize / per-byte compare loop far more time...	depth_reach	depth_lift, side_lift, winner_deeper	depth_lift = winner_line_count / max(loser_line_count, 1) >> 1 AND winner_deeper = true: the VPC-resolving corpus drives the gate line through many more executions per seed than the cmplog-blocking corpus. DISCRIMINATING: a pure-I2S (final-compare-only) alternative predicts depth_lift ~ 1.0 AND winn...	depth_lift >= 2.0 AND winner_deeper == true
vp_drives_parser_depth_into_structured_true_side	LW_L (vp/cmp+naive)	1	for token-CLASS / structural-ENTRY gates with NO single counted literal, value_profile's gradient admits a corpus that PARSES DEEPER: its resolving seeds drive the gate line's TRUE side substantially more per-seed than the naive...	depth_reach	depth_lift, side_lift, winner_deeper	winner_deeper = true AND side_lift >= 2.0. DISCRIMINATING vs token-presence: depth_reach measures ACTUAL execution into the structured branch. DISCRIMINATING vs naive-reaches-it: cards report naive true-side T=0.	winner_deeper == true AND side_lift >= 2.0
vp_gradient_structural_depth_assembly	L_WL (vpc/cmp+naive)	1	vpc's CMP_MAP gradient accumulates partial credit across chained comparisons to drive the corpus to a structurally DEEPER parse state (multi-child content model / namespace binding / internal subset / pointer-identity) whose gate...	depth_reach	w_depth, l_depth, depth_lift	W seeds reach a structurally DEEPER parse state than L seeds (depth_lift > 1, w_depth meaningfully above l_depth), because the gradient banked credit on each chained comparison to climb the multi-token structure. This is DISCRIMINATING: the competing 'literal-token climb' alternative (sibling hypot...	depth_lift >= 1.5 AND w_depth > l_depth
[VP-pro] vp_admits_structurally_richer_corpus (final feature, n=5)							
vp_admits_token_dense_corpus	_WWL (vp+vpc/naive)	3	value_profile's CMP_MAP gradient admits a token-DENSE corpus: the VP-resolving seed set carries far more of the gate's required grammar/structural tokens PER BYTE than the naive-blocking set, because the gradient rewards concentr...	struct_size_and_token_count	token_density_lift, vp.token_density, naive.token_dens...	token_density_lift >= 2.0 (VP corpus carries at least 2x more grammar tokens per byte than naive). DISCRIMINATING vs the refuted scalar-distance story (density counts whole-token PRESENCE per byte, orthogonal to residual Hamming distance), vs the falsified bigness story (here VP seeds are SMALLER),...	token_density_lift >= 2.0 AND vp.token_density > naive.token_density
vp_admits_larger_valid_structure	_WWL (vp+vpc/naive)	2	For STRUCTURAL FORMAT-INTEGRITY gates (sfnt/font assembly), value_profile's gradient admits a corpus of PHYSICALLY LARGER, valid-header seeds than the naive-blocking corpus – the gradient toward well-formed table directories pro...	joint_necessity	size_lift, tag_lift, winner_tags, loser_tags	size_lift >= 1.3 AND tag_lift >= 1.0 : the VP-resolving font corpus is >=30% larger AND carries >=1 more valid sfnt table tag than the naive-blocking corpus. DISCRIMINATING: a no-gradient null predicts size_lift ~ 1.0 AND tag_lift ~ 0; the SQL/grammar branches FALSIFY the bigness direction (VP smal...	size_lift >= 1.3 AND tag_lift >= 1.0
[VP-pro] vp_operand_byte_enrichment (final feature, n=5)							
vp_enriches_fixed_offset_fourcc_operand	LW_L (vp/cmp+naive)	1	for the fixed-offset FOURCC gate (lcms_2141 DeviceClass 'scnr' at offset 0x0c-0x0f, decoy 'mntr'), value_profile's gradient enriches the resolving corpus for the TARGET operand bytes relative to the DECOY bytes at the gate offset...	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich > 0 AND signed_target_enrich > decoy_enrich. DISCRIMINATING vs naive luck: a no-gradient corpus shows signed_target_enrich ~ 0. DISCRIMINATING vs pure-I2S: cmplog (I2S, no gradient) still BLOCKS (Layer-1).	signed_target_enrich > 0 AND signed_target_enrich > decoy_enrich
vp_enriches_gate_operand_bytes	LWW_ (vp+vpc/cmp)	1	secondary / corroborant: for the exact-codepoint conjunction gates, value_profile's gradient enriches the VPC corpus for the target operand bytes (vs decoy bytes) at the gate site, because near-miss codepoints register as new cov...	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich > 0 (and > decoy_enrich): the VPC-resolving corpus is enriched for the target codepoint bytes relative to decoy bytes at the gate site. Discriminating: a pure-I2S corpus shows signed_target_enrich ~ 0 (I2S substitutes the operand on demand without persistently enriching the sav...	signed_target_enrich > 0 AND signed_target_enrich > decoy_enrich

Raw cluster	Shape (res/blk)	#br	Cluster description	Tool	Metric(s)	Metric meaning / prediction	Verify rule
vp_enriches_size_tag_bytes	L_W_ (vpc/cmp)	1	value_profile's CMP_MAP gradient enriches the VPC-resolving corpus for the target literal bytes relative to decoy bytes at the gate offset: near-miss byte matches register as new coverage, so the gradient persistently steers the ...	operand_enrichment	signed_target_enrich, decoy_enrich	signed_target_enrich > 0 AND signed_target_enrich > decoy_enrich: the VPC-resolving corpus is enriched for the target literal bytes relative to decoy bytes at the gate offset. DISCRIMINATING: cmplog carries the IDENTICAL I2S mutator, so a pure-I2S byte-substitution advantage leaves no persistently ...	signed_target_enrich > 0 AND signed_target_enrich > decoy_enrich
vp_gradient_fixed_offset_operand_enrichment	LWWL (vp+vpc/cmp+naive)	1	value_profile's CMP_MAP gradient ENRICHES the resolving corpus, at the fixed gate offset, with bytes equal to (or progressively closer to) the target FOURCC operand, relative to the decoy class the losers carry – a fixed-offset ...	operand_enrichment	signed_target_enrich, decoy_enrich	G1, discriminating: if VP's gradient ratchets the DeviceClass dword from 'mntr' toward 'prtr', value_profile's saved corpus is byte-enriched, at offsets 0x0c-0x0f, toward the target FOURCC relative to naive (signed_target_enrich > 0) while NOT enriched toward the decoy (decoy_enrich <= 0). ALTERNAT...	signed_target_enrich >= 1.0 AND decoy_enrich <= 0.0
vp_gradient_fixed_offset_byte_enrichment	L_WL (vpc/cmp+naive)	1	vpc's CMP_MAP gradient enriches the saved corpus with the target byte at a FIXED file offset on a single-byte enumerated-constant gate, where a Hamming-distance climb is degenerate (1-byte distance is binary) but a corpus byte-pr...	operand_enrichment	signed_target_enrich, decoy_enrich	The resolving corpus is ENRICHED for the target byte 0x03 at offset 0x19 (signed_target_enrich > 0) while the decoy 0x06 is NOT enriched there. This is DISCRIMINATING: the competing 'I2S substitution alone suffices' alternative predicts cmplog (I2S, no gradient) would enrich the target byte just as...	signed_target_enrich >= 1.0 AND decoy_enrich < signed_target_enrich
[JOINT (I2S×VP)] joint_value_distance_closure (final feature, n=18)							
joint_value_precision	LLW_ (vpc/cmp+vp)	9	vpc and cmplog reach a SIMILARLY-sized input (size_lift ~ 1), but vpc lands a specific gradient-climbed VALUE (a single exact multi-byte keyword/token) that the cmplog-blocking seeds approach byte-by-byte yet never reach; I2S is ...	value_distance_reached	vpc_min_distance, cmp_min_distance, distance_closure_r...	Inputs are the SAME size (size_lift ~ 1.0 – a 'PRAGMA <name>;' statement is no bigger than the SELECT/junk blocking fragments) BUT the vpc-resolving seeds reach a keyword value strictly CLOSER to the exact target token than the cmplog-blocking seeds ever do: cmp_min_distance > 0 (cmplog seeds pref...	size_lift < 1.3 AND cmp_min_distance > 0 AND distance_closure_ratio >= 0.5
joint_value_precision	LLWL (vpc/cmp+vp+naive)	6	vpc and cmplog reach a SIMILARLY-sized structure (size_lift ~ 1, sometimes vpc smaller), but vpc places a specific gradient-climbed VALUE that the cmplog-blocking seeds approach yet never reach; I2S is still necessary to place th...	value_distance_reached	vpc_min_distance, cmp_min_distance, distance_closure_r...	Structures are the SAME size (size_lift ~ 1.0, the assembly-depth proxy is blind here) BUT the vpc-resolving seeds reach a value strictly CLOSER to the target operand than the cmplog-blocking seeds ever do: cmp_min_distance > 0 (cmplog seeds approach but never hit the exact value) while vpc_min_dis...	size_lift < 1.3 AND cmp_min_distance > 0 AND distance_closure_ratio >= 0.5
joint_fixed_offset_literal_presence	LLW_ (vpc/cmp+vp)	2	NEW (round 2, KEEP). The decisive joint divergence is GATED BY A FIXED-OFFSET CONTIGUOUS MULTI-BYTE LITERAL (a FOURCC table tag at a constant low input offset) where the BLOCKING corpus carries JUNK/absence at that offset: the va...	operand_enrichment	signed_target_enrich, decoy_enrich	At the fixed tag offset the RESOLVING corpus carries the exact literal far more than the BLOCKING corpus does: signed_target_enrich > 0 (the value_profile-blocking corpus is depleted of the exact FOURCC – it holds JUNK/absence decoy bytes – while the I2S-bearing resolving corpus concentrates the ...	signed_target_enrich >= 1.0

Raw cluster	Shape (res/blk)	#br	Cluster description	Tool	Metric(s)	Metric meaning / prediction	Verify rule
joint_value_ gradient_closure	LLWL (vpc/cmp+vp+naive)	1	NEW (round 1). The cmplog-BLOCKING arm reaches a value that is STRICTLY CLOSER to the gate operand than naive/value-profile but still NON-zero, while the vpc-RESOLVING arm lands it (distance ~ 0): the value-profile CMP_MAP gradie...	value_ distance_ reached	winner_closer, distance_gap, distance_ closure_ratio, c...	The cmplog-blocking arm gets CLOSER to the operand than the gradient-less baselines yet stalls at cmp_min.distance > 0; the vpc-resolving arm closes it (vpc_min.distance ~ 0), so winner_closer == true and distance_gap > 0 with a substantial distance_closure_ratio. Discriminating: a PURE-I2S story p...	winner_closer == true AND cmp_min.distance > 0 AND distance_closure_ratio >= 0.34
[JOINT (I2SxVP)] joint_assembly_depth (final feature, n=9)							
joint_assembly_ depth	LLWL (vpc/cmp+vp+naive)	6	vpc assembles a LARGER, multi-field structure than I2S-only (cmplog) ever reaches; the gradient is necessary to grow size/token-count past where exact substitution stalls, and I2S is necessary because the gradient alone (value.pr...	joint_ necessity	vp_tags, size_ lift	On the value-profile-BLOCKING seed set the structural-token count is ~0 (vp_tags ~ 0): the gradient alone cannot place ANY exact multi-byte structural token, so I2S is necessary. AND the vpc-RESOLVING median seed is materially BIGGER than the cmplog-BLOCKING median seed (size_lift >= 1.3): the grad...	vp_tags < 0.3 AND size_lift >= 1.3
joint_gradient_ iteration_depth	LLWL (vpc/cmp+vp+naive)	2	NEW (round 3). The gradient's necessity is visible as ITERATION/EXECUTION DEPTH at the decisive branch line: the vpc-RESOLVING corpus drives the gate line markedly MORE times per seed (and actually takes the decisive side) than t...	depth_reach	depth_lift, side_lift, winner_deeper	The vpc-resolving corpus drives the decisive branch LINE markedly deeper per seed than the cmplog-blocking corpus: depth_lift >= 1.5 AND winner_deeper == true (the gradient retains the near-miss inputs that re-enter and iterate the gated region; the cards show vpc line-counts an order of magnitude ...	depth_lift >= 1.5 AND winner_deeper == true
joint_assembly_ depth	LLW_ (vpc/cmp+vp)	1	vpc assembles a LARGER, multi-field structure than I2S-only (cmplog) ever reaches; the gradient is necessary to grow size/token-count past where exact substitution stalls, and I2S is necessary because the gradient alone (value.pr...	joint_ necessity	vp_tags, size_ lift	On the value-profile-BLOCKING seed set the structural-token count is ~0 (vp_tags ~ 0): the gradient alone cannot place ANY exact multi-byte structural token (FOURCC table tag, ICC colorspace signature, XML markup literal), so I2S is necessary. AND the vpc-RESOLVING median seed is materially BIGGER ...	vp_tags < 0.3 AND size_lift >= 1.3
[ctx-coverage] ctx_iteration_path_depth (final feature, n=14)							
iteration_path_ depth_reach	ctx_coverage_ WL	14	naive_ctx's resolving corpus drives the blocker line DEEPER per seed (more loop/recursion/path re-entries, or actually takes the blocked side) than naive's blocking corpus	depth_reach	depth_lift, winner_ deeper, side_ lift	naive_ctx's saved corpus drives the blocker line to a strictly higher per-seed execution count than naive's (depth_lift >= 2.0 AND winner_deeper == True). A value/byte-solving mechanism mis-bucketed into this naive_ctx>naive shape would reach the blocker line at the SAME depth and differ only in op...	depth_lift >= 2.0 AND winner_deeper == True
[ctx-coverage] ctx_corpus_inflation (final feature, n=2)							
ctx_corpus_inflation	ctx_coverage_ LW	2	ctx inflates the corpus / map so naive_ctx hoards far more (context-redundant) entries than naive	corpus_size_ ratio	corpus_count_ ratio	corpus_count_ratio >= 1.5 (naive_ctx hoards >=1.5x the corpus naive keeps, because the same edges count as new under each call-context). A NON-ctx penalty (e.g. naive simply got luckier on byte mutation) would predict corpus_count_ratio ~ 1.0 - no systematic corpus bloat on the blocking arm.	corpus_count_ratio >= 1.5
[ctx-coverage] ctx_depth_inflation (final feature, n=2)							

Raw cluster	Shape (res/blk)	#br	Cluster description	Tool	Metric(s)	Metric meaning / prediction	Verify rule
ctx_depth_ inflation_at_gate	ctx_coverage_ LW	2	a high-call-count / multi-context path at the gate is what ctx fragments (flooding) vs a shallow exact-literal gate (dilution)	depth_reach	call_count_lift	FLOODING members (deep high-iteration loop, e.g. libpng inflate loop) show call_count_lift >> 1 (naive_ctx re-enters the loop function far more, manufacturing the (ctx,edge) flood); DILUTION members (shallow 1-hop exact-literal gate, e.g. harfbuzz.6124) show call_count_lift ~ 1 - the gate is shall...	call_count_lift >= 3.0 (flooding sub-mechanism) vs call_count_lift < 3.0 (dilution sub-mechanism)
[ngram-coverage] ngram_sequential_depth_reach (final feature, n=2)							
ngram_sequential_ depth_reach	ngram_ coverage_ WL	2	n-gram path-tuple retention lets naive_ngram4 reach deeper sequential-decode path states that plain edge coverage collapses and discards	depth_reach	winner_ depth, loser_ depth, depth_ lift	The naive_ngram4-resolving (winner) seed set reaches a strictly deeper sequential-decode path state at the blocker than the naive-blocking (loser) seed set: winner.depth > loser.depth (depth_lift >= 1.5). This is discriminating because an alternative where naive simply lacks the right LITERAL bytes...	depth_lift >= 1.5 AND winner_depth > loser_depth
[grimoire] grimoire_structural_token_assembly (final feature, n=9)							
structural_token_ assembly	grimoire_ structural_WL	7	grimoire's grammar mutators place structural tokens/constructs into the corpus that naive's byte havoc cannot, building larger/deeper inputs	joint_ necessity	naive_tags, grimoire_tags, tag_lift, size_ lift	naive_tags ~ 0 (byte havoc places essentially no coherent structural tokens) AND grimoire_tags >> naive_tags AND size_lift >= 1.0. A pure-byte-mutation alternative predicts naive_tags ~ grimoire_tags and tag_lift ~ 0.	tag_lift >= 1.0 AND naive_tags < grimoire_tags AND grimoire_tags >= 2
structural_token_ richness	grimoire_ structural_WL	2	grimoire's GeneralizationStage + structural splice/recurse mutators place MORE DISTINCT coherent structural tokens/constructs per seed than cmplog's I2S byte-substitution can, even though cmplog can place individual literals	joint_ necessity	grimoire_tags, cmplog_tags, tag_lift, size_ lift	grimoire_tags - cmplog_tags >= 1.0: a grimoire seed carries on average at least one MORE distinct sfnt-structural token than a cmplog blocking seed. Rules out BYTE-RICHNESS-NOT-STRUCTURE: cmplog runs I2S, so if the advantage were merely placing magic/tag LITERALS (a byte capability cmplog ALSO has ...	tag_lift >= 1.0 AND grimoire_tags >= 2 AND size_lift >= 1.0
[grimoire] grimoire_structural_size_depth (final feature, n=2)							
structural_size_ depth_only	grimoire_ structural_WL	2	fallback: grimoire reaches the gate via larger/deeper structure (state-machine depth, repeated category blocks, nested clusters) without a countable target-vocabulary tag signal	joint_ necessity	size_lift	size_lift >= 1.3 (grimoire seeds materially larger/deeper) even when tag_lift is ~0, because the gate is about state-machine DEPTH / repeated category structure rather than a countable table tag.	size_lift >= 1.3

Mechanism families — detailed evidence

One table per technique-direction family. Each row is a final mechanism with a representative validated branch, its real-corpus measured metrics, the deterministic verify rule that fired, and the mechanism reading.

I2S-pro (238 roadblocks, 4 mechanisms)

Mechanism (n; shapes)	Example branch & gate	Real-corpus metrics	Verify rule (arbiter)	Description / reading
i2s_string_literal_substitution (n=185) LLW_(vpc/cmp+vp); WLWL (cmp+vpc/vp+naive); WLW_(cmp+vpc/vp); WL_L (cmp/vp+naive); ...	curl/118 – operand=HTTP/, gate=inequality_or_range	winner_literal_count=1; presence_ratio=0.333; winner=cmplog; loser=naive; get=curl	literal_winner_literal_count >= 1.0 AND literal_presence_ratio <= 0.34	I2S splices one or more logged CMP anchor literals (URL scheme '://', 'HTTP'/'HTTP/2', header names, protocol tokens 'pop3', chunk-type FOURCCs 'IDAT'/'sRGB'/'eXif'/'sCAL'/'tIME', OpenType table tags 'GPOS'/'GSUB', the Mach-O magic, the switch comparand byte) somewhere into the input; the literal is PRESENT in the cmplog-resolving corpus but ABSENT (deplet...
i2s_numeric_tag_substitution (n=39) WLWL (cmp+vpc/vp+naive); WL_L (cmp/vp+naive); W_WL (cmp+vpc/naive); _LWL (vpc/vp+nai...	lcms/2110 – operand=2, gate=switch_case	signed_target_enrich=18.002	signed_target_enrich >= 0.8	cmplog (I2S) substitutes a logged CMP literal at a fixed gate offset, so its corpus is enriched in the target value at that offset relative to naive; value_profile/naive cannot synthesize the exact literal
i2s_structural_assembly_reach_depth (n=8) WLWL (cmp+vpc/vp+naive); W_WL (cmp+vpc/naive)	lcms/2113 – operand=5, gate=switch_case	depth_lift=14125.0; side_lift=6.32; winner_deeper=True; target=lcms	depth_lift >= 2.0 AND side_lift >= 3.0 AND winner_deeper == true	I2S assembles a MULTI-FIELD structure whose downstream effect is that the cmplog-RESOLVING corpus, run dynamically, DRIVES execution deeper to the blocker line AND TAKES the decisive side, while the naive/value_profile-BLOCKING corpus reaches the line only shallowly and never takes that side. Discriminator = ASSEMBLED-STRUCTURE EXECUTION DEPTH, not the pres...
i2s_operand_value_precision (n=6) WLWL (cmp+vpc/vp+naive)	curl/129 – derived gate	distance_gap=0.8; distance_closure_ratio=1.0; winner_closer=True; target=curl	winner_closer == true AND distance_gap > 0 AND distance_closure_ratio >= 0.5	NEW (round 3). The blocking arm (naive AND, critically, value_profile — which has the CMP-MAP feedback but NO operand-substitution mutator) REACHES the gate line just as deeply as the winner (often deeper: the cards show losers hitting the dispatcher/switch/predicate line many times — libxml2 L hits=390, bloaty FindContaining L=1720, lcms_2330 losers satura...

I2S-anti (49 roadblocks, 3 mechanisms)

Mechanism (n; shapes)	Example branch & gate	Real-corpus metrics	Verify rule (arbiter)	Description / reading
i2s_anti_target_depletion (n=25) LW_W (vp+naive/cmp); L_W (naive/cmp)	libpng/3977 operand=PNG_FIXED_ ERROR, gate=equality	signed_target_enrich=-0.821	skip != 'no_gate_signature' AND signed_target_enrich < -0.3	I2S splices a WRONG/competing but abundant CMP operand into the input (a well-known font-table tag, a script-tag/codepoint constant, an in-range fixed-point magnitude, or a competing protocol value), pinning the cmplog corpus onto that decoy and DEPLETING it of the target byte value at the winner gate's offset; naive/value_profile never harvest the decoy so...

Mechanism (n; shapes)	Example branch & gate	Real-corpus metrics	Verify rule (arbiter)	Description / reading
i2s_anti_decoy_overfit (n=14) LWLW (vp+naive/cmp+vpc); L_LW (naive/cmp+vpc); - WLW (vp+naive/vpc)	harfbuzz/5958 – derived gate	distance_gap=0.5; closer=True; target=harfbuzz	winner_closer == True and distance_gap >= 0.15	Gate is an EXACT-equality test against a short codepoint literal that the WINNER (non-I2S) corpus lands by free havoc byte mutation, while the I2S stage diverts mutation energy away from the literal so the I2S-arm corpus stalls FARTHER from it in Hamming distance
i2s_anti_structural_byte_corruption (n=10) LWWW (vp+vpc+naive/cmp)	harfbuzz/5831 – derived gate	signed_target_enrich=-13.644	signed_target_enrich < -0.3	I2S RandReplace overwrites the exact structural/category/literal bytes naive havoc assembles, depleting (not enriching) the target byte in cmplog’s corpus relative to naive

VP-pro (53 roadblocks, 4 mechanisms)

Mechanism (n; shapes)	Example branch & gate	Real-corpus metrics	Verify rule (arbiter)	Description / reading
vp_gradient_value_distance_closure (n=28) LLWL (vpc/cmp+vp+naive); LLW_(vpc/cmp+vp); LWLL (vp/cmp+vpc+naive); LWWL (vp+vpc/cm...	lcms/2087 – derived gate	size_lift=0.43; cmp_min. distance=0.5; distance_closure. ratio=1.0; target=lcms	size_lift < 1.3 AND cmp_min. distance > 0 AND distance. closure_ratio >= 0.5	vpc and cmplog reach a SIMILARLY-sized structure (size_lift ~ 1, sometimes vpc smaller), but vpc places a specific gradient-climbed VALUE that the cmplog-blocking seeds approach yet never reach; I2S is still necessary to place the surrounding exact literal(s), and the gradient is necessary for the last-mile value precision the structural proxy is blind to
vp_gradient_drives_assembly_depth (n=15) LLWL (vpc/cmp+vp+naive); LWLL (vp/cmp+vpc+naive); LWWL (vp+vpc/cmp+naive); LWW_(vp+...	harfbuzz/5616 – derived gate	depth_lift=157.2; winner. deeper=True; target=harfbuzz	depth_lift >= 1.5 AND winner. deeper == true	NEW (round 3). The gradient’s necessity is visible as ITERATION/EXECUTION DEPTH at the decisive branch line: the vpc-RESOLVING corpus drives the gate line markedly MORE times per seed (and actually takes the decisive side) than the cmplog-BLOCKING corpus, which reaches only the gate’s vicinity and fires the line a handful of times before stalling one compar...
vp_admits_structurally_richer_corpus (n=5) _WWL (vp+vpc/naive)	harfbuzz/5302 – derived gate	tag_lift=1; size_lift=4.69; tar- get=harfbuzz	size_lift >= 1.3 AND tag_lift >= 1.0	For STRUCTURAL FORMAT-INTEGRITY gates (sfnt/font assembly), value_profile’s gradient admits a corpus of PHYSICALLY LARGER, valid-header seeds than the naive-blocking corpus – the gradient toward well-formed table directories produces bigger, more-tokened inputs. (CARRIED FORWARD; validated round-2 for harfbuzz/5302.)
vp_operand_byte_enrichment (n=5) LWWL (vp+vpc/cmp+naive); LWW_(vp+vpc/cmp); LW_L (vp/cmp+naive); L- WL (vpc/cmp+naive...	lcms/2143 – operand=prtr, gate=switch_case	signed_target_enrich=17.85; decoy. enrich=-0.895	signed_target_enrich >= 1.0 AND decoy_enrich <= 0.0	value_profile’s CMP_MAP gradient ENRICHES the resolving corpus, at the fixed gate offset, with bytes equal to (or progressively closer to) the target FOURCC operand, relative to the decoy class the losers carry – a fixed-offset value-enrichment signal, distinct from the sliding-window min-distance the value_distance test uses

JOINT (I2SxVP) (27 roadblocks, 2 mechanisms)

Mechanism (n; shapes)	Example branch & gate	Real-corpus metrics	Verify rule (arbiter)	Description / reading
joint_value_distance_closure (n=18) LLWL (vpc/cmp+vp+naive); LLW_(vpc/cmp+vp)	lcms/2087 – derived gate	size_lift=0.43; cmp_min_distance=0.5; distance_closure_ratio=1; target=lcms	size_lift < 1.3 AND cmp_min_distance > 0 AND distance_closure_ratio >= 0.5	vpc and cmplog reach a SIMILARLY-sized input (size_lift ~ 1), but vpc lands a specific gradient-climbed VALUE (a single exact multi-byte keyword/token) that the cmplog-blocking seeds approach byte-by-byte yet never reach; I2S is ...
joint_assembly_depth (n=9) LLWL (vpc/cmp+vp+naive); LLW_(vpc/cmp+vp)	harfbuzz/5399 – derived gate	vp_tags=0; size_lift=2.08; target=harfbuzz	vp_tags < 0.3 AND size_lift >= 1.3	vpc assembles a LARGER, multi-field structure than I2S-only (cmplog) ever reaches; the gradient is necessary to grow size/token-count past where exact substitution stalls, and I2S is necessary because the gradient alone (value_profile) places ~0 structural tokens

ctx-coverage (18 roadblocks, 3 mechanisms)

Mechanism (n; shapes)	Example branch & gate	Real-corpus metrics	Verify rule (arbiter)	Description / reading
ctx_iteration_path_depth (n=14) ctx.coverage_LW; ctx.coverage_WL	libxml2/6576 – operand=0x0d, gate=equality	depth_lift=4.39; deeper=True; target=libxml2	depth_lift >= 2.0 and winner_deeper == True	a high-call-count / multi-context path at the gate is what ctx fragments (flooding) vs a shallow exact-literal gate (dilution)
ctx_corpus_inflation (n=2) ctx.coverage_LW	lcms/2149 – operand=1, gate=inequality_or_range	corpus_count_ratio=1.722; winner=naive_ctx; loser=naive; target=lcms	corpus_count_ratio >= 1.5	ctx inflates the corpus / map so naive_ctx hoards far more (context-redundant) entries than naive
ctx_depth_inflation (n=2) ctx.coverage_LW	libxml2/6576 – derived gate	depth_lift=4.39; deeper=True; target=libxml2	depth_lift >= 2.0 and winner_deeper == True	a high-call-count / multi-context path at the gate is what ctx fragments (flooding) vs a shallow exact-literal gate (dilution)

ngram-coverage (2 roadblocks, 1 mechanisms)

Mechanism (n; shapes)	Example branch & gate	Real-corpus metrics	Verify rule (arbiter)	Description / reading
ngram_sequential_depth_reach (n=2) ngram.coverage_WL	libxml2/6325 – derived gate	winner_depth=156.67; depth=95.45; target=libxml2; loser_depth_lift=1.64	depth_lift >= 1.5 AND winner_depth > loser_depth	n-gram path-tuple retention lets naive_ngram4 reach deeper sequential-decode path states that plain edge coverage collapses and discards

grimoire (11 roadblocks, 2 mechanisms)

Mechanism (n; shapes)	Example branch & gate	Real-corpus metrics	Verify rule (arbiter)	Description / reading
grimoire_structural_token_assembly (n=9) grimoire_structural_WL	libxml2/6924 – derived gate	tag_lift=4.1; grimoire_tags=6.1; naive_tags=2; target=libxml2	tag_lift >= 1.0 AND naive_tags < grimoire_tags AND grimoire_tags >= 2	grimoire's grammar mutators place structural tokens/constructs into the corpus that naive's byte havoc cannot, building larger/deeper inputs
grimoire_structural_size_depth (n=2) grimoire_structural_WL	libxml2/6743 – gate=equality	size_lift=2.22; target=libxml2	size_lift >= 1.3	fallback: grimoire reaches the gate via larger/deeper structure (state-machine depth, repeated category blocks, nested clusters) without a countable target-vocabulary tag signal